WILL ROCHA

PhD Astrophysics
Data Scientist / ML Expert

CONTACT & INFO

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🔗 linkedin.com/in/will-rocha
🔗 GitHub - ENIIGMA project
🔗 GitHub - LIDA project
🌐 My website
🛂 **Permanent Residence**

TECHNICAL STACK

Programming & SQL
Python, SQL, CUDA, OpenMP, NumPy, SciPy, JavaScript

AI & Generative AI
PyTorch, scikit-learn, VAE, GAN, DDPM (Diffusion Models), U-Nets

Data Engineering & Cloud
ETL Pipelines, Relational Databases, Data Harmonization, Databricks, MLflow

Tools & MLOps
Git, Docker, Binder, Linux (Bash), Jupyter, Colab, Kaggle

Domain Expertise
Hyperspectral Data, Optical Sensing, FTIR/UV-vis Spectroscopy, Monte Carlo

EDUCATION

PhD in Astrophysics
Vale do Paraíba University (2015)

LANGUAGES

English: Fluent
Portuguese: Native
Dutch: A2 (Inburgeringsdiploma). B1 is coming!

> **Summary:** Senior Data Scientist with 10+ years of experience engineering end-to-end data pipelines, ML frameworks, and scientific software for high-dimensional hyperspectral and clinical datasets. Proven track record leading cross-functional international teams (5–20 members) and turning complex sensor data into scalable data platforms adopted by NASA, ESA, and Airbus.

WORK EXPERIENCE

Molecular Spectroscopy Scientist — Radboud University — **Apr 2026 – Present**

- Built an end-to-end Deep Learning PyTorch pipeline to quantify 10 breath gases from FTIR spectroscopy data.
- Developed a GenAI pipeline using Diffusion Models (DDPM) to synthesize missing datasets for training robustness.
- Implemented attention-enhanced U-Nets for spectral baseline estimation with physics-informed loss functions.

AI & Medical Imaging Data Scientist — Upwork / Mercor — **Mar 2025 – Mar 2026**

- Delivered a PyTorch pipeline (U-Net & attention architectures) for brain-MRI anomaly detection, reaching 95% precision across 3 classes.
- Engineered physics-informed light-scattering simulation algorithms, translating first-principles physics into production code.

Senior Scientist & Data Lead — Leiden University — **Feb 2021 – Jan 2025**

- **Big Data Analytics:** Harmonized and analyzed terabyte-scale hyperspectral data from the James Webb Space Telescope, using statistical inference and pattern recognition to uncover key astrochemistry insights.
- **Data Engineering (LIDA Platform):** Architected and deployed the LIDA platform featuring a relational SQL backend, automated ETL pipelines, and interactive Bokeh frontends—scaled to become the world's largest infrared spectroscopy database used by NASA and ESA.
- **Leadership & Strategy:** Directed international teams of 5–15 scientists and software developers, aligning project roadmaps, stakeholder requirements, and technical data deliverables.
- **Technical Documentation:** Authored 50+ high-impact technical papers and reports (including *Nature Astronomy*), featured in official NASA and ESA global press releases.

Scientific Software Developer — University of Copenhagen — **Sep 2018 – Jan 2021**

- Designed and deployed ENIIGMA, an open ML framework for automated feature extraction, genetic modeling algorithms, and pattern recognition on high-dimensional scientific data.

COMPUTATIONAL MODELING & ALGORITHM ENGINEERING

- **Numerical Optimization & Signal Processing (NKABS Code):** Engineered custom optical modeling algorithms implementing Kramers-Kronig transformations to solve complex refractive index equations; applied advanced signal processing and numerical optimization to extract material opacities from high-dimensional spectra.
- **High-Performance Computing & Monte Carlo Simulations (ProDiMo & RADMC-3D):** Executed large-scale 3D radiative transfer and thermochemical modeling; leveraged Monte Carlo algorithms and parallelized computing workflows to simulate complex physical systems and physics-informed Data Science tasks.

TECHNICAL CONSULTANCY & ADVISORY SERVICES

Lead Technical Consultant & Project Advisor — Leiden Univ. – TU Delft — **2023 – 2025**

- Provided technical advisory, payload specification guidance, and data workflow consulting in collaboration with major industry partners:
  - **Airbus (Methane Constellation):** Consulted on spatial/spectral resolution constraints and processing workflows for global CH4 monitoring.
  - **European Space Agency (ESA):** Advised on scientific trade-offs, instrument selection, and payload requirements for the Iapetus Mission.
  - **ACRI-ST:** Guided the development of algorithmic spectral extraction pipelines for Earth observation datasets.